

Santiago Martínez Novoa

Systems and Computing Engineer

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RESEARCH INTERESTS

My research focuses on multilingual NLP and multimodal systems that work for low-resource and underrepresented languages, with an emphasis on cross-lingual transfer and rigorous, culturally grounded evaluation. In parallel, I build production data and machine learning systems for higher education, from learning analytics to student-success prediction.

EDUCATION

Universidad de los Andes

M.Sc. Information Engineering

GPA: 4.80/5.00

Bogotá, Colombia

2025 – 2026 (*To be conferred*)

Universidad de los Andes

B.Sc. Systems and Computing Engineering

GPA: 4.58/5.00, *Cum Laude*

Minor in French

Bogotá, Colombia

2021 – 2024

PUBLICATIONS

Nguyen, Q. P., Gaschi, F., Anugraha, D., **Martínez Novoa, S.**, Lee, E.-S. A. “Online Language Adaptive Sampling for Better Distributed Cross-lingual Gains.” *Findings of EMNLP 2026, Budapest, Hungary* (2026, to appear).

Martínez Novoa, S., Melo Barrero, F., Ruiz Giraldo, M., Velásquez Marín, S., Carvajal Chaves, N., Hernández Hoyos, M., Reyes, J. P. “Centralizing the Student Lens: Design and Evaluation of a Learning Analytics Dashboard for Academic Advising in Higher Education.” *WEEF (IFEES & GEDC) 2026, Cartagena de Indias, Colombia* (2026, to appear).

Martínez Novoa, S., Gómez Mesa, L., Prieto, J., Manrique, R. “MeSHClass-ES and AnatEM-ES: Open Resources for Spanish Biomedical NLP.” *BioNLP 2026 (ACL Workshop), San Diego, California*, pp. 617–629 (2026). DOI: [10.18653/v1/2026.bionlp-1.49](https://doi.org/10.18653/v1/2026.bionlp-1.49).

Soto Parada, E. G., **Martínez Novoa, S.**, Klopstock Triana, N., Fuquen Florez, D. A., Peña, F. C., Hernández Hoyos, M., Reyes, J. P. “No Student Left in Limbo: Multi-Level Clustering for Success Prediction in a Latin American University.” *AIED 2026, WideAIED Track, Communications in Computer and Information Science, Springer*, pp. 63–77 (2026). DOI: [10.1007/978-3-032-29791-4_5](https://doi.org/10.1007/978-3-032-29791-4_5).

Martínez Novoa, S., Ibáñez, M. C., Gómez Mesa, L., Kramer, J. “Classification of Chest XRay Diseases through image processing and analysis techniques.” (2026). [arXiv:2601.10913](https://arxiv.org/abs/2601.10913).

Martínez Novoa, S., Rozo Fajardo, N., González Vargas, D. A., Bedoya Figueroa, N. “Efficient Multilingual Dialogue Processing via Translation Pipelines and Distilled Language Models.” (2026). [arXiv:2601.09059](https://arxiv.org/abs/2601.09059).

Martínez Novoa, S. “Detection and Segmentation of Malignant Melanoma Regions in Dermoscopic Images Using Machine Learning.” *Undergraduate thesis, Universidad de los Andes* (2024). Available at: [Uniandes Repository](#).

RESEARCH EXPERIENCE

Lee Language Lab, Ontario Tech University

2026-05 – Present

Mitacs Globalink Research Intern - Collaborating Researcher

Oshawa, Canada

- Co-authored a study on online adaptive language sampling for better distributed cross-lingual transfer, accepted to Findings of EMNLP 2026.
- Lead its direct continuation, studying when to stop cross-lingual realignment: more realignment is not always better for transfer, and low-resource languages offer no labeled data to tell when to stop; paper under review at ACL Rolling Review (ARR).
- Co-develop M2-ALIGN, studying where multilingual alignment should live in vision-language models: how to extend visual grounding to 11 languages, including low-resource ones, without eroding the multilingual reasoning the base models already have; ongoing work on culturally grounded evaluation (CVQA).
- Contributed to a PRISMA systematic literature review of multilingual, multimodal, and multicultural conversational agents.

FLAG Research Lab - A.I. and Machine Learning, Universidad de los Andes

2025-01 – Present

Evaluation and Enhancement of Natural Language Models in spanish

Bogotá, Colombia

- Research on improving natural language model capabilities for Spanish medical contexts, focusing on text classification and named entity extraction tasks under Professor Ruben Manrique Ph.D.
- Develop and train specialized models from scratch for specific Spanish medical domain tasks.
- Conduct comparative evaluations between highly-trained models and Large Language Models (LLMs) using prompting techniques such as zero-shot and few-shot learning.

Augmented Artificial Intelligence Initiative, Universidad de los Andes

2025-01 – 2025-06

Generative Artificial Intelligence for Research - Research Assistant

Bogotá, Colombia

- Focused on optimizing university research processes through custom AI agent development and automated evaluation systems.
- Built locally-deployed RAG systems using Ollama models and integrated Gemini API calls to answer domain-specific research questions.
- Designed LLM evaluation architecture to assess response quality and accuracy across different research contexts using Prometheus, BERT score and OpenAI API.
- Automated student assessment processes by developing LLM-based evaluation systems that grade responses against established rubrics and expected answers.

College of Computing and Information Science, Cornell University

2024-06 – 2024-08

Summer Research Intern

Ithaca, United States

- Designed a Data Discovery architecture to explore and modify datasets in various formats under Professor Sainyam Galhotra Ph.D.
- Developed the first version of the architecture including the upload and scoring through semantic search of multiple databases.

WORK EXPERIENCE

EdTech Node, Universidad de los Andes

2026-03 – Present

Data Scientist

Bogotá, Colombia

- Manage the unit's data end to end, from source systems to published products: the institutional information behind the academic advising platform and university-wide data products such as academic percentiles and student dropout indicators.
- Lead the unit's adoption of institutional data governance and quality policies: establishing data ownership and origin with source units, building column-level definitions, and defining quality rules to audit, correct,

and monitor its data products.

- Design, validate, and deploy machine learning models to predict student dropout and academic risk.
- Lead the team's research line, which has produced two peer-reviewed publications (AIED 2026 and WEEF 2026) and a journal article in preparation.

Vice Dean's Office of Student Affairs, Universidad de los Andes

2025-01 – 2026-03

Data Scientist - Graduate Research Assistant

Bogotá, Colombia

- Processed academic data and developed Power BI dashboards under the guidance of Professor Marcela Hernández Hoyos, Ph.D. and Professor Juan Pablo Reyes Gómez, Ph.D., delivering actionable insights on student performance and institutional metrics.
- Optimized ETL pipeline performance by 87% (reduced processing time from 15 to 2 minutes) by migrating from Pandas to Polars for data retrieval and storage from Azure Blob Storage.
- Enabled the university to save approximately \$100K USD annually by replacing an external academic counseling platform.
- Co-implemented a risk detection system that alerts stakeholders about over 1000 critical student cases.
- Designed and implemented the unit's complete data architecture in Databricks, following a medallion architecture and fully automating data processing and institutional information storage pipelines.

Precia PPV S.A

2024-08 – 2024-12

Machine Learning Intern

Bogotá, Colombia

- Co-develop and train a Multi-layer Perceptron (MLP) to reduce the calculation time of a critical financial metric by 70%, providing clients with significantly faster access to preliminary estimates.
- Design and implement end-to-end AWS infrastructure including robust ETL pipelines integrating multiple data sources (databases, FSx servers) and automate model deployment using AWS Glue and Lambda functions.
- Conduct data exploration and engineering to optimize model accuracy and efficiency across development and production environments.
- Presented our work at the Analytics Forum 2025 on model optimization and infrastructure architecture, including the use of MLPs, AWS pipelines, and data transformation strategies.

Vice Dean's Office of Student Affairs, Universidad de los Andes

2024-08 – 2024-12

Data Scientist - Undergraduate Research Assistant

Bogotá, Colombia

- Co-implemented the ETL process (initially in Pandas, later migrated to Polars) to retrieve data from Azure Blob Storage, structure it, and store it in a relational database.
- Developed and conducted integrity tests to ensure the coherence and accuracy of a multidimensional model, validating against the original data.

Digital Transformation, Universidad de los Andes

2023-09 – 2024-05

Backend Developer

Bogotá, Colombia

- Applied expertise in Azure Functions and Python to actively contribute to software development projects, particularly focusing on seamless interaction with the Open AI API.
- Created comprehensive documentation for the project's infrastructure, emphasizing clarity and transparency. This initiative facilitated smoother collaboration among team members and enhanced the overall understanding of the software architecture.

Research Center of the Faculty of Engineering, Universidad de los Andes

2023-08 – 2023-12

Frontend Developer

Bogotá, Colombia

- Co-developed and implemented responsive and user-friendly frontend solutions for a software project at the university using Angular.
- Implemented dynamic access controls, tailoring user permissions based on roles and responsibilities.

Digital Transformation, Universidad de los Andes

2023-06 – 2023-08

Cross-Cultural Liaison and Software Developer

Bogotá, Colombia

- Facilitated effective communication between directors and 5 exchange students by providing translation services.
- Collaborated on software development projects utilizing Open AI and Azure Functions, successfully integrating the developed software into an interactive chatbot.

TEACHING EXPERIENCE

Universidad de los Andes

2025-06 – Present

Graduate Teaching Assistant (LLM-Based Applications & Agent Development)

Bogotá, Colombia

- Graduate TA across three semesters for two sequential 8-week master's courses, 'LLM-Based Applications' (RAG systems) and 'Agent Development', supporting over 200 master's students.
- Created tutorials and course materials for the RAG systems module using LangChain, LangGraph, and LangSmith, covering retrieval techniques, indexing, and evaluation with RAGAS.
- Created tutorials and course materials for the agent development module using LangGraph and LangSmith, covering agent design, tool use, and multi-agent workflow orchestration.
- Evaluated student performance on practical projects and implementation exercises throughout both courses.

Systems and Computing Engineering Department, Universidad de los Andes

2024-01 – 2024-12

Undergraduate Research Assistant (CupiTaller Programming Support Center)

Bogotá, Colombia

- Supervised approximately 40 tutors and 70 teaching assistants as part of the 7-person team overseeing operations at CupiTaller, the university's programming support center, under Professor Eduardo Rosales, Ph.D.
- Designed programming projects for 'Introducción a la Programación', a course with +1100 students each semester.
- Cleaned, enriched, and processed datasets essential for each project, ensuring data accuracy completeness and consistency.

Systems and Computing Engineering Department, Universidad de los Andes

2021-08 – 2023-08

Undergraduate Teaching Assistant - Programming Tutor (Programming Support Center)

Bogotá, Colombia

- Provided 5 hours of weekly Python programming tutoring at the University's programming help center.
- Instructed more than 100 students, helping them develop fundamental coding skills in Python.

Universidad de los Andes

2022-01 – 2022-12

Undergraduate Teaching Assistant (Introduction to Programming)

Bogotá, Colombia

- Provided support to the professor during programming teaching laboratories, assisting approximately 40 students each semester.
- Corrected and graded basic programming projects.

Universidad de los Andes

2023-01 – 2023-06

Undergraduate Teaching Assistant (Fundamentals of Technological Infrastructure)

Bogotá, Colombia

- Corrected and graded tests, activities, and assessments
- Guided approximately 40 students to a thorough understanding of Fundamentals of Technological Infrastructure

HONORS AND AWARDS

Mitacs Globalink Research Internship

2026-05

Ontario Tech University

- Competitive international Mitacs award funding a 12-week research internship at the Lee Language Lab, Ontario Tech University, under the supervision of Professor Annie En-Shiun Lee, Ph.D.

Cum Laude

2025-04

Universidad de los Andes

- A Cum Laude degree is awarded to students with a GPA in the highest 3% GPA of all the students that have graduated from Universidad de Los Andes in a given faculty in the last five years.

Summer Undergraduate Research Fellowship (SURF)

2024-06

Cornell University

- Financial support awarded to participate in the Summer Undergraduate Research Fellowship (SURF) program at Cornell University

INVITED TALKS

Accelerating Credit Value Adjustment with Machine Learning. *Analytics Forum 2025 - Universidad de los Andes*, Bogotá, Colombia. 2025-05-09. [\[Web page\]](#)

SKILLS

Languages

- Spanish (Native)
- English (Advanced)
- French (Conversational)

Technical Skills

- *Programming Languages*: Python, SQL, JavaScript, TypeScript
- *Machine Learning & AI*: Scikit-learn, Keras, PyTorch, TensorFlow
- *LLM & Natural Language Processing*: Transformers, BERT, LangChain, LangGraph, LangSmith, Ollama
- *Data Processing*: Pandas, Polars, Databricks
- *Cloud & DevOps*: AWS, Azure Functions, Azure Blob Storage
- *Databases*: PostgreSQL, MongoDB, ChromaDB, Pinecone